UF UNIVERSITY of FLORIDA

Department of Mathematics

MAP 2302
Exam 1A
Spring 2023

A. Sign your bubble sheet on the back at the bottom in ink.

B. In pencil, write and encode in the spaces indicated:

1) Name (last name, first initial, middle initial)

2) UF ID number

3) Section number

C. Under "special codes" code in the test ID numbers 1, 1.

- 2 3 4 5 6 7 8 9 0
- 2 3 4 5 6 7 8 9 0

D. At the top right of your answer sheet, for "Test Form Code", encode A.

- B C D E

E. 1) This test consists of 13 multiple choice questions of five points in value, plus 3 free response questions worth 35 points.

2) The time allowed is 100 minutes.

3) You may write on the test.

4) Raise your hand if you need more scratch paper or if you have a problem with your test. DO NOT LEAVE YOUR SEAT UNLESS YOU ARE FINISHED WITH THE TEST.

F. KEEP YOUR BUBBLE SHEET COVERED AT ALL TIMES.

G. 1) Before turning in your test **check carefully for transcribing errors**. Any mistakes you leave in are there to stay.

2) You must turn in your scantron and tearoff sheets to your exam proctor. Be prepared to show your picture I.D. with a legible signature.

3) The answers will be posted in Canvas shortly. Your instructor will return your tearoff sheets with your exam score in lecture. Your score will also be posted in Canvas within one week of the exam.

4) Scantron errors in name, UF ID, section number, and form code will result in a five-point deduction.

NOTE: Be sure to bubble the answers to questions 1−13 on your scantron.

Questions 1 − 13 are worth 5 points each.

1. Solve $\dfrac{dy}{dx} = \sqrt{x+y} - 1$ using an appropriate substitution.

a. $2\sqrt{x+y+1} = x + C$ b. $2\sqrt{x-y} = y + C$ c. $2\sqrt{x-y} = x + C$

d. $2\sqrt{x+y} = x + C$ e. $\sqrt{x+y} = y + C$

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2. Which of the following satisfies the differential equation $\dfrac{d^2y}{dx^2} + y = 0$?

a. $y = \cos 2t + \sin 5t$ b. $y = 2\cos t + 5\sin t$ c. $y = 2e^t$

d. $y = x^2 + 1$ e. $y = \arctan t$

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3. Solve the initial value problem.

$$y' = \frac{y}{x} + xe^x, \ y(-1) = 0$$

a. $y = xe^x - ex$ b. $y = xe^x + ex$ c. $y = xe^x - \dfrac{x}{e}$

d. $y = e^{-x} - e^x$ e. $y = x^2e^x$

4. Which of the following statements best describes the differential equation

$$(y^3 + 5x^2y)dx + (x^3 + xy^2)dy = 0?$$

a. This equation is exact.

b. This equation can become exact by multiplying by an integrating factor $\mu(x)$ that is a function of x alone.

c. This equation can become exact by multiplying by an integrating factor $\mu(y)$ that is a function of y alone.

d. This equation is linear.

e. This equation is separable.

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5. Where does the Theorem of Existence and Uniqueness imply the existence of a unique solution in the $ty-$plane for the differential equation below?

$$\frac{dy}{dt} = \frac{1}{y} + \sqrt{y-t};\ y(t_0) = y_0$$

a. $(t_0, y_0) = (5, -1)$

b. $(t_0, y_0) = (-4, 0)$

c. $(t_0, y_0) = (7, 7)$

d. $(t_0, y_0) = (-3, -2)$

e. None of these

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6. The half-life of carbon-14 is approximately 5,730 years. A sample has a mass of 200g of carbon-14. Find the time in years it takes for the mass to be reduced to 50g.

a. $5730\frac{\ln(1/2)}{\ln(1/4)}$

b. $\frac{\ln(1/4)}{5730\ln(1/2)}$

c. $\frac{\ln(1/2)}{5730\ln(1/2)}$

d. $5730\frac{\ln(1/4)}{\ln(1/2)}$

e. $\frac{\ln(5730\cdot 1/4)}{\ln(1/2)}$

7. Which differential equation below has the highest order?

a. $u\dfrac{\partial^2 u}{\partial x^2} = t\dfrac{\partial u}{\partial t} + x\dfrac{\partial u}{\partial x}$

b. $\ln(y^5) = 4x\dfrac{d^3y}{dx^3}$

c. $t\dfrac{d^4y}{dt^4} = 4y + t^2$

d. $\dfrac{\partial^3 v}{\partial^2 x \partial t} = v^{10} + x^8 - v_{tt}$

e. $y''' - 3x^2y'' + 3y^4y' - 2y = x^7$

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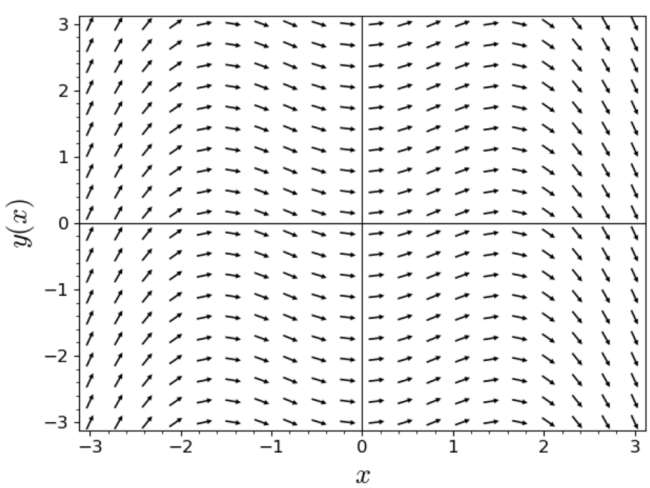
8. The direction field pictured below is described by the differential equation



a. $\dfrac{dy}{dx} = x\sin x$ b. $\dfrac{dy}{dx} = x\cos x$ c. $\dfrac{dy}{dx} = \sin x$

d. $\dfrac{dy}{dx} = y\cos x$ e. $\dfrac{dy}{dx} = y\sin x$

9. Using an appropriate substitution, find the general solution to the homogeneous differential equation. (Hint: You may need to simplify your answer to obtain one of the answer choices.)

$$\frac{dy}{dx} = \frac{y[\ln(y) - \ln(x)]}{x}.$$

a. $y = xe^{Cx+1}$

b. $y = xe^{Ce^x}$

c. $y = e^{1+Cx}$

d. $y = e^{Ce^x}$

e. $y = 0$

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10. For which value(s) of k is $\frac{dx}{dt} + x^k = t^{k+2}$ linear?

a. -2 only
b. 0 only
c. 1 only
d. 0 or -2
e. 0 or 1

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11. Find an integrating factor to make the DE exact.

$$(x + y)\sin(y)dx + (x\sin(y) + 1)dy = 0$$

a. $\cot(y)$

b. $\csc(y)$

c. $\sin(y) + \cos(y)$

d. $\tan(y)$

e. $\sec(y)$

12. Which of the following equations is homogenous?

a. $(x + 3y + 3)dx + (7x - 3y)dy = 0$

b. $x \sin(\frac{y}{x})\frac{dy}{dx} = y \sin(\frac{y}{x}) + x$

c. $\frac{d^2x}{dy^2} + 10\frac{dy}{dx} - 3xy = \cos(x)$

d. $\frac{dy}{dx} + \sin(x)y = x^3$

e. $\frac{dy}{dx} - 7y = 4x$

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13. Solve the Bernoulli equation $\dfrac{dy}{dx} = \dfrac{2y}{x} - x^2y^2$.

a. $y = \dfrac{5x^2}{x^5 + Cx}$

b. $y = \dfrac{x^2}{x^5 + C}$

c. $y = \dfrac{5y^2}{x^5 + C}$

d. $y = \dfrac{y^2}{x^5 + C}$

e. $y = \dfrac{5x^2}{x^5 + C}$

MAP 2302 Exam 1A, Part II Free Response

Name: _____ Section #: _____

SHOW ALL WORK TO RECEIVE FULL CREDIT

1. (13 pts)

a. Solve the first order linear initial value problem $y' - \dfrac{y}{x} = \dfrac{x}{x^2+1}$; $y(1) = 0$. (9 pts)

b. Find $\lim\limits_{x\to\infty} y(x)$ (4 pts)

2. (10 pts) Suppose we are using Euler's method to approximate the solution to the initial value problem $\dfrac{dy}{dx} = y^2 - 1;\ y(0) = 0.$

a. Let h be the step size used, with $h > 0$. If, after two iterations, we get that $y_2 = 0$, what is h? Use exact forms if possible. (6 pts)

b. Using the step size you found in part a., compute y_3 and y_4. Show your work. Use exact forms if possible. (4 pts)

3. (12 pts) Given the exact DE and initial condition $\dfrac{dy}{dx} = \dfrac{5x^4 - xy^2}{4y^3 + x^2y}$; $y(0) = 1$,

a. Solve the IVP. (9 pts)

b. The solution to the IVP intersects the line $y = 1$ for two values of x. One of those values is $x = 0$. What is the other one? (3 pts)

University of Florida Honor Pledge:

On my honor, I have neither given nor received unauthorized aid doing this exam.

Signature: _____